

DSA Day 12: Stack (array & linked)

Book: Data Structures Through C in Depth - Ch4 §4.1 · pp 108-113 | Code in Python

Overview

A stack is LIFO. Push/pop/peek are $O(1)$. Implement with a Python list or with linked nodes.

Key concepts to master

- top pointer
- overflow/underflow
- LIFO order
- Array vs linked implementation

Python implementation

```
class Stack:
    def __init__(self): self.a=[]
    def push(self,x): self.a.append(x)
    def pop(self): return self.a.pop() if self.a else None
    def peek(self): return self.a[-1] if self.a else None
```

Complexity

push/pop/peek $O(1)$.

Common pitfalls

- Popping an empty stack
- Fixed-size array overflow

Practice

- Push 1,2,3 then pop all
- Give 3 real uses of a stack

